Scalability & Future Plan

Date	03 October 2026
Team ID	TEAM-ID-TO-BE-FILLED
Project Name	PocketSmart AI: Your Smart Budget & Recommendation Assistant
Maximum Marks	1 Mark

Current System Limitations

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Sessions and token blacklist are in memory	Users are signed out on restart; not shareable across several server processes	High
2	Shopping links are search URLs and prices are AI estimates	Prices may differ from live prices	High
3	SQLite and a single Uvicorn worker	Limits write concurrency and horizontal scaling	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single worker; 25 concurrent users tested with 0 % errors (offline mode)	Several Uvicorn/Gunicorn workers behind Nginx; container autoscaling
2	Data Storage	SQLite file	PostgreSQL with connection pooling; object storage (S3) for outfit images
3	Performance	Gemini call is synchronous per request, run in a threadpool	Cache repeated requests, use async Gemini client, background queue for long jobs
4	Security	bcrypt, JWT in HttpOnly cookie, upload validation	Redis session store and token revocation, rate limiting, HTTPS-only cookies, secrets manager

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Live product data via retailer APIs, price-drop alerts	1-2 months	More accurate prices and deeper personalization
Phase 3	Hosted deployment (Docker on Render/Railway), CI with GitHub Actions	2-3 months	Public demo URL and automated testing
Phase 4	New planners (travel, groceries), Google sign-in, PDF export, regional languages	3-6 months	Wider audience and retention